μ's Paper 3

Time allowed: 1 hour 15 minutes

This is a practice Paper 3 for all AA HL students. It is probably harder than what will come up on your final examination, but nevertheless it should be good practice. Enjoy!

Instructions

- Answer all questions.
- Show all necessary working clearly.
- Unless otherwise stated, exact values are preferred.
- A graphic display calculator is required for this paper.

May 2026

1. [Maximum mark: 29]

The following question investigates a family of curves defined by a constant product of distances from two fixed points.

Consider two fixed points $A(-a, 0)$ and $B(a, 0)$, where $a \in \mathbb{R}^+$.

A point $P(x, y)$ moves so that the product of its distances from A and B is constant:

$$\left|\overrightarrow{PA}\right| \cdot \left|\overrightarrow{PB}\right| = b^2,$$

where $b > 0$.

The curve traced out by P is called a *Cassini oval*. This family of curves has many applications in physics as it is linked to the lines of equipotential around massive bodies.

(a) **(i)** Write the equation of the curve traced by P. **[2]**

(ii) Hence show that

$$\left((x+a)^2 + y^2\right)\left((x-a)^2 + y^2\right) = b^4.$$ **[1]**

(b) **(i)** By expanding the left-hand side, show that

$$\left((x+a)^2 + y^2\right)\left((x-a)^2 + y^2\right) = (x^2 + y^2 + a^2)^2 - 4a^2x^2.$$

(ii) Hence deduce that the equation of the curve can be written as

$$(x^2 + y^2 + a^2)^2 - 4a^2x^2 = b^4.$$ **[4]**

(c) Using the equation from **(b) (ii)** explain why the curve is symmetric about both the x-axis and the y-axis. **[2]**

(This question continues on the following page.)

(d) **(i)** Show that the x-intercepts satisfy

$$|x^2 - a^2| = b^2.$$

(ii) Hence find all possible x-intercepts.

(iii) Show that the intersections with the y-axis exist only when

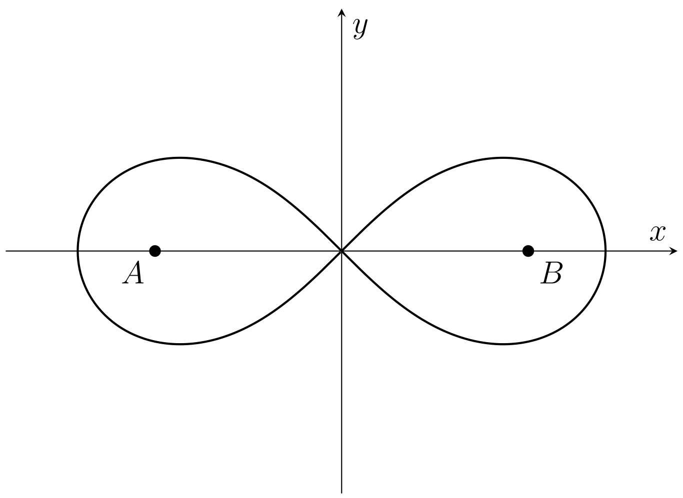
$$b \geq a.$$ **[6]**

(e) Now consider the case $b = a$.

Show that the equation of the curve becomes

$$(x^2 + y^2)^2 = 2a^2(x^2 - y^2).$$ **[4]**

This curve is called the *lemniscate of Bernoulli*.



Sketch of the lemniscate of Bernoulli

(f) Using

$$x = r\cos\theta, \qquad y = r\sin\theta,$$

show that the curve can be written

$$r^2 = 2a^2\cos(2\theta).$$ **[3]**

(g) For the right-hand loop of the lemniscate,

$$-\frac{\pi}{4} \leq \theta \leq \frac{\pi}{4}.$$

By writing the expression for the curve in **(f)** as $y^2 = f(\theta)$, where $f(\theta)$ is a function to be determined, or otherwise, find the maximum value of y on the right-hand loop. **[7]**

2. [Maximum mark: 26]

This question investigates a sequence of nested radicals and shows how it leads to a general infinite product formula.

Consider the sequence (a_n) defined by

$$a_0 = 0, \qquad a_{n+1} = \sqrt{2 + a_n}, \qquad n \geq 0.$$

(a) Find a_1, a_2 and a_3, giving your answers in exact form. **[2]**

(b) **(i)** Show that, for $0 \leq \theta \leq \pi$,

$$\sqrt{2 + 2\cos\theta} = 2\cos\left(\frac{\theta}{2}\right).$$

[2]

(ii) Hence prove the following statement by induction for all integers $n \geq 0$:

$$a_n = 2\cos\left(\frac{\pi}{2^{n+1}}\right).$$

[6]

(c) We now define the sequence $(P_n)_{n\in\mathbb{N}}$ as follows:

$$P_n = \prod_{k=1}^{n} \frac{a_k}{2},$$

where $\prod$ denotes product notation:

$$\prod_{i=1}^{n} u_i = u_1 \cdot u_2 \cdot u_3 \cdots u_n.$$

(i) Show that

$$P_n = \prod_{k=1}^{n} \cos\left(\frac{\pi}{2^{k+1}}\right).$$

[1]

(ii) By using the identity

$$\sin(2\theta) = 2\sin\theta\cos\theta$$

repeatedly or otherwise, show that

$$1 = 2^n \sin\left(\frac{\pi}{2^{n+1}}\right) P_n.$$

[5]

(This question continues on the following page.)

(iii) Hence show that

$$P_n = \frac{1}{2^n \sin\left(\frac{\pi}{2^{n+1}}\right)}.$$

Using the standard limit

$$\lim_{u \to 0} \frac{\sin u}{u} = 1 \quad \text{or} \quad \sin(u) \approx u \quad \text{as} \quad u \to 0,$$

or otherwise, deduce that

$$\lim_{n \to \infty} P_n = \frac{2}{\pi}.$$

[3]

This gives the infinite product

$$\frac{2}{\pi} = \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{2+\sqrt{2}}}{2} \cdot \frac{\sqrt{2+\sqrt{2+\sqrt{2}}}}{2} \cdots .$$

(d) The result above can be generalised.

For $0 < x < \pi$, we define a new sequence $(b_n(x))$ by

$$b_0(x) = 2\cos x, \qquad b_{n+1}(x) = \sqrt{2 + b_n(x)}.$$

You are given that

$$b_n(x) = 2\cos\left(\frac{x}{2^n}\right)$$

for all integers $n \geq 0$.

We now introduce

$$Q_n(x) = \prod_{k=1}^{n} \frac{b_k(x)}{2}.$$

(i) Show that

$$Q_n(x) = \frac{\sin x}{2^n \sin\left(\frac{x}{2^n}\right)}.$$

[4]

(ii) Hence deduce that

$$\prod_{k=1}^{\infty} \cos\left(\frac{x}{2^k}\right) = \frac{\sin x}{x}$$

for $0 < x < \pi$.

[3]